

MedTrack DV

Quality Assurance (QA) Testing Document

Hospital Operations & Patient Analytics Dashboard

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Tableau Desktop / Tableau Public | Source: Fact_Admissions_Clean.csv

1. Executive Summary

This report presents the final quality assurance validation of the MedTrack DV Hospital Operations & Patient Analytics Tableau solution. The objective of the QA exercise was to independently reconcile all 11 implemented KPI values against the supplied source datasets and the KPI values displayed in Tableau, while also checking aggregation logic, rounding, population integrity and source traceability.

The final validation covered 11 KPIs. All 11 KPI values reconcile to the Tableau values under the documented calculation rules. The final QA result is therefore PASS, with no KPI calculation mismatches identified.

QA Control	Final Result
KPI measures tested	11
KPIs independently validated	11 / 11
KPI calculation mismatches	0
Total admissions population	45,000
Unique patients	23,275
Readmission records	1,120 Yes / 43,880 No
Final KPI QA status	PASS

2. QA Objective

The QA objective is to establish that the KPI values presented in Tableau are accurate, reproducible from the appropriate source data, and protected against common BI errors such as incorrect aggregation, duplicate counting, inappropriate joins and rounding inconsistencies.

- Validate every implemented KPI rather than sampling only selected KPIs.
- Recalculate each KPI independently from its appropriate source dataset.
- Compare independent results with the Tableau KPI values.
- Check distinct-count logic for patients, departments, wards, beds and doctors.
- Control repeated ward-capacity values to avoid double counting.
- Validate readmission totals and the resulting readmission percentage.
- Document the calculation method, expected value, Tableau value and final status.

3. QA Scope

Scope Area	Coverage
Source Data Validation	Fact_Admissions_Clean.csv, Dim_Doctor.csv, Readmission_Summary.csv
KPI Validation	All 11 implemented KPI measures
Aggregation Validation	COUNT, COUNTD, AVG and derived measures
Reconciliation	Source result vs Tableau KPI
Rounding Validation	Percentage and decimal display rules
Data Integrity	Population totals and category reconciliation
Exception Handling	Mismatch/source-traceability assessment

4. Source Data & Evidence

The project workflow defines the Admission table as the primary fact table because it connects patients, departments, wards, beds and diseases. The project also documents the creation of Fact_Admissions_Clean.csv and the separate handling of one-to-many datasets to prevent duplicated admission-level metrics.

The project specification identifies KPI engineering, dashboard development, testing and validation as formal project stages, with KPI calculation validation and healthcare metric verification required before delivery.

Evidence Source	QA Use
Fact_Admissions_Clean.csv	Admissions, patients, departments, wards, beds and length-of-stay validation
Dim_Doctor.csv	Independent validation of Total Doctors
Readmission_Summary.csv	Independent validation of readmission counts and rate
Tableau KPI Sheets / Dashboard	Expected displayed KPI values

5. KPI Validation Methodology

Each KPI was tested using a source-to-dashboard reconciliation method:

1. Identify the KPI definition and business entity being measured.
2. Identify the authoritative source field(s).
3. Apply the appropriate aggregation or formula independently.
4. Compare the independent result with the Tableau KPI.
5. Calculate or assess any variance and rounding difference.
6. Review the result for duplication or incorrect aggregation risk.
7. Record PASS only when the value and calculation logic reconcile.

This approach is consistent with the project requirement to validate KPI calculations and verify healthcare metrics before final delivery. The project target specifies KPI accuracy above 95%.

6. Final KPI Reconciliation Matrix

Test ID	KPI	Validation Formula / Logic	Independent	Tableau	Variance	Status
QA-KPI-01	Total Patients	COUNTD(Patient ID)	23,275	23,275	+0	PASS
QA-KPI-02	Total Admissions	COUNT(Admission ID) / admission-row population	45,000	45,000	+0	PASS
QA-KPI-03	Total Departments	COUNTD(Department)	6	6	+0	PASS
QA-KPI-04	Total Wards	COUNTD(Ward)	27	27	+0	PASS
QA-KPI-05	Total Beds	SUM one capacity value per distinct ward	415	415	+0	PASS
QA-KPI-06	Total Doctors	COUNTD(Doctor ID)	98	98	+0	PASS
QA-KPI-07	Available Beds	COUNTD(Bed Number)	145	145	+0	PASS
QA-KPI-08	Occupied Beds	Total Beds - Available Beds	270	270	+0	PASS
QA-KPI-09	Bed Occupancy Rate	Occupied Beds / Total Beds × 100	65.06%	65%	+0.06 pp (rounding)	PASS
QA-KPI-10	Average Length of Stay	AVG(Length of Stay)	5.155 days	5.16 days	-0.005 days (rounding)	PASS
QA-KPI-11	Readmission Rate	Readmissions / Total Readmission Summary × 100	2.49%	2.49%	-0.00 pp	PASS

7. Detailed KPI Test Results

QA-KPI-01 — Total Patients

Validation logic: COUNTD(Patient ID).

Independent evidence: 23,275 unique patients were identified in the fact dataset.

Tableau value: 23,275 | Status: PASS.

Analyst assessment: Independent distinct-patient count exactly matches Tableau.

QA-KPI-02 — Total Admissions

Validation logic: COUNT(Admission ID) / admission-row population.

Independent evidence: 45,000 admission records were validated.

Tableau value: 45,000 | Status: PASS.

Analyst assessment: Admission population exactly matches Tableau.

QA-KPI-03 — Total Departments

Validation logic: COUNTD(Department).

Independent evidence: 6 distinct departments were identified.

Tableau value: 6 | Status: PASS.

Analyst assessment: Distinct department count exactly matches Tableau.

QA-KPI-04 — Total Wards

Validation logic: COUNTD(Ward).

Independent evidence: 27 distinct wards were identified.

Tableau value: 27 | Status: PASS.

Analyst assessment: Distinct ward count exactly matches Tableau.

QA-KPI-05 — Total Beds

Validation logic: Sum one capacity value per distinct ward.

Independent evidence: 415 total beds after de-duplicating repeated ward capacity values.

Tableau value: 415 | Status: PASS.

Analyst assessment: Capacity is aggregated at ward level to avoid repeated admission-row inflation.

QA-KPI-06 — Total Doctors

Validation logic: COUNTD(Doctor ID).

Independent evidence: 98 unique doctors were identified in Dim_Doctor.csv.

Tableau value: 98 | Status: PASS.

Analyst assessment: Independent doctor-source count exactly matches Tableau.

QA-KPI-07 — Available Beds

Validation logic: COUNTD(Bed Number).

Independent evidence: 145 distinct bed numbers were identified.

Tableau value: 145 | Status: PASS.

Analyst assessment: Distinct-bed count matches Tableau.

QA-KPI-08 — Occupied Beds

Validation logic: Total Beds - Available Beds.

Independent evidence: $415 - 145 = 270$.

Tableau value: 270 | Status: PASS.

Analyst assessment: Derived occupied-bed value exactly matches Tableau.

QA-KPI-09 — Bed Occupancy Rate

Validation logic: $\text{Occupied Beds} / \text{Total Beds} \times 100$.

Independent evidence: $270 / 415 \times 100 = 65.06\%$.

Tableau value: 65% | Status: PASS.

Analyst assessment: Tableau rounds 65.06% to the displayed whole-number value of 65%.

QA-KPI-10 — Average Length of Stay

Validation logic: $\text{AVG}(\text{Length of Stay})$.

Independent evidence: Independent average = 5.155 days.

Tableau value: 5.16 days | Status: PASS.

Analyst assessment: 5.155 days rounds to Tableau's 5.16 days.

QA-KPI-11 — Readmission Rate

Validation logic: $\text{Readmissions} / \text{Total Readmission Records} \times 100$.

Independent evidence: $1,120 / 45,000 \times 100 = 2.49\%$.

Tableau value: 2.49% | Status: PASS.

Analyst assessment: Readmission Summary reconciles to 1,120 Yes and 43,880 No; total = 45,000.

8. Data Integrity & Aggregation Controls

- Patient IDs were evaluated as distinct entities; they were not summed as numeric values.
- Department and ward KPIs use distinct counts to avoid duplicate counting across admission rows.
- Ward capacity was aggregated once per distinct ward before calculating Total Beds.
- Available Beds uses distinct Bed Number values rather than raw admission-row counts.
- Occupied Beds is reconciled as Total Beds minus Available Beds.
- Average Length of Stay uses AVG rather than SUM.
- Readmission categories reconcile to the full 45,000-record population.
- Percentage KPIs are checked at full precision and then compared with the Tableau display after rounding.

9. Defect & Exception Assessment

Assessment Area	Finding	Severity	Disposition
KPI calculation mismatch	No mismatches identified across 11 KPIs.	None	Closed
Aggregation risk	Ward capacity can repeat across admission rows if aggregated incorrectly.	Low	Controlled through distinct-ward aggregation
Patient ID aggregation	SUM(Patient ID) would produce an invalid measure.	High	Controlled; KPI uses COUNTD
Bed calculation	Repeated ward/bed rows can inflate capacity if raw rows are counted.	Medium	Controlled through distinct entity logic
Readmission source	Readmission Summary now provides source-level Yes/No counts.	None	Fully validated
Doctor source	Dim_Doctor provides unique doctor IDs.	None	Fully validated

10. QA Test Case Summary

Category	Planned / Tested	Passed	Failed	Status
KPI Validation	11 / 11	11	0	PASS
Population Reconciliation	Admission population and readmission population	Passed	0	PASS
Distinct Entity Validation	Patient / Department / Ward / Bed / Doctor	Passed	0	PASS
Derived KPI Validation	Occupied Beds / Occupancy Rate	Passed	0	PASS
Rounding Validation	Occupancy Rate / Average LOS	Passed	0	PASS
Readmission Validation	Yes / No counts and rate	Passed	0	PASS

11. QA Acceptance Criteria

- All 11 implemented KPIs reconcile to their appropriate source data or documented calculation.
- No unexplained KPI variance remains.
- No KPI is affected by avoidable duplicate counting.
- Distinct-count measures use the correct business entity.
- Derived measures use transparent formulas.
- Rounding differences are documented and do not represent calculation errors.
- Readmission and doctor KPIs are supported by their dedicated source datasets.
- The final QA result meets the project's KPI validation objective.

12. Final QA Assessment

The final QA exercise successfully validated all 11 KPI measures implemented in the MedTrack DV Tableau solution. Independent source recalculations matched the Tableau values for Total Patients, Total Admissions, Total Departments, Total Wards, Total Beds, Total Doctors, Available Beds and Occupied Beds. Average Length of Stay reconciled at 5.155 days and displayed correctly as 5.16 days. Bed Occupancy Rate reconciled at 65.06% and displayed correctly as 65% after rounding.

The Readmission Rate was fully validated using the dedicated Readmission Summary: 1,120 Yes and 43,880 No records out of 45,000, producing 2.49%, exactly matching the Tableau KPI of 2.49%.

Based on the source-to-dashboard reconciliation performed for this report, all 11 implemented KPIs have passed QA validation. No KPI calculation defect remains open. The KPI layer is therefore suitable to proceed to final dashboard testing evidence and project submission.

Sign-Off Item	Result
KPI QA execution	Completed
Source reconciliation	Completed
Calculation mismatch review	Completed — 0 mismatches
Readmission validation	Completed
Doctor KPI validation	Completed
Overall QA recommendation	APPROVED FOR FINAL SUBMISSION

13. Project Context & Traceability

The project specification defines the solution as a hospital operations and patient analytics dashboard suite with four interconnected dashboards—Hospital Overview, Patient Flow, Department Analytics and Resource Utilization. It also identifies testing and validation as a formal module and specifies QA Checklist and Dashboard Testing Report deliverables.